

# Weather Systems & the Water Cycle

**Use the cue, then fade the cue.** Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

|                  |                                                                                                                                                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cue focus</b> | Solar energy drives evaporation; water vapor condenses into clouds, falls as precipitation, and returns through runoff, infiltration, and collection before cycling again. |
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## Round 1 - Retrieve It

1. What process changes liquid water to water vapor?

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2. What process changes water vapor to liquid droplets?

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3. What is precipitation?

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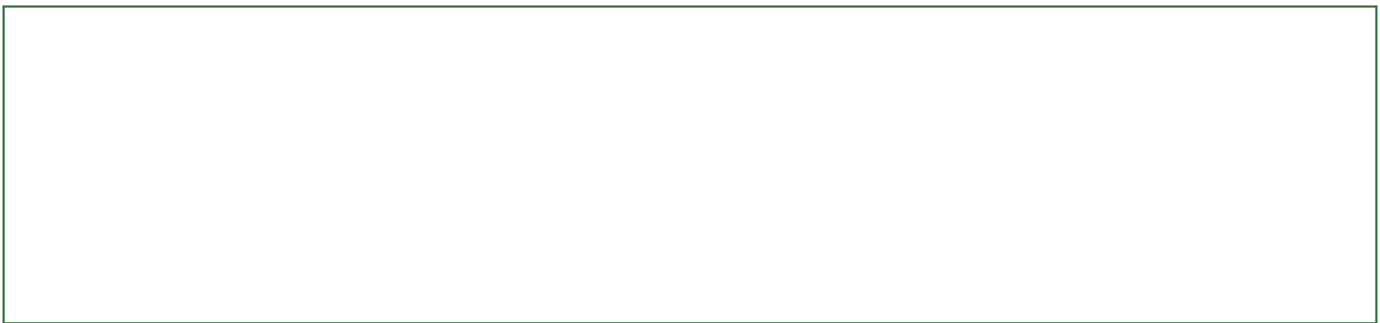
4. What major energy source drives the water cycle?

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## Round 2 - Sketch It

Draw a water-cycle diagram with evaporation, condensation, precipitation, runoff, and collection. Add the Sun as the energy source.



# Weather Systems & the Water Cycle - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

## Round 3 - Apply the Relationship

1. A puddle becomes smaller on a warm day. Which process dominates?

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2. Water droplets form on the outside of a cold cup. Which process?

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3. Rainwater flows over land into a creek. Which process is being described?

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## Round 4 - Reason from Evidence

Trace one water molecule from the ocean into a cloud, onto land, and back toward the ocean. Name the processes along the way.

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# Weather Systems & the Water Cycle - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

**A****Articulate**

Explain Weather Systems & the Water Cycle in your own words. Use one example that is not printed on the cue card.

**C****Connect**

Connect the water cycle to weather. Why are condensation and cloud formation important to precipitation?

**E****Extend**

Predict how several very hot, dry days could affect evaporation from a shallow pond and the pond's water level.

**Confidence check** Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

# TEACHER KEY - Weather Systems & the Water Cycle

**Remove this page before student copying.** Answers below give expected ideas, not required wording.

## Retrieve It

1. Evaporation.
2. Condensation.
3. Water that falls from clouds, such as rain, snow, sleet, or hail.
4. The Sun.

## Sketch It - look for

Correct process order/relationships, with water moving between surface and atmosphere; Sun linked to evaporation.

## Apply It

1. Evaporation.
2. Condensation of water vapor from air.
3. Runoff.

## Reason from Evidence - look for

Example: evaporation -> condensation -> precipitation -> runoff/groundwater/collection -> ocean.  
Transpiration/infiltration can be included if accurately used.

## ACE - Extend look-for

Evaporation rate likely increases, lowering pond level if inputs do not replace the lost water.

### Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.